VIT-AP UNIVERSITY

Amaravati, Andhra Pradesh, India

School of Computer Science and Engineering

Designing Test Cases Report

FITPULSE

Systematic Validation Plan for Auth, Data, UI, and Deployment

Course	Software Engineering
Assessment Component	6 - Designing Test Cases
Maximum Marks	10
Allocated Marks	10
Submission Date	14 April 2026
Test Scope	Functional, integration, auth security, state flow, and release validation
Live Production Link	https://fitpulse-swe.vercel.app/
GitHub Repository Link	https://github.com/abhinavsai2006/fitpulse

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1. Introduction

This report defines comprehensive test cases for the latest FitPulse baseline, including Firebase Authentication, Firestore profile persistence, environment-variable based configuration, and production deployment behavior.

1.1 Test objectives

- Validate all critical user journeys from authentication to dashboard usage.
- Verify verified-email policy, password reset, and Google sign-in fallback behavior.
- Ensure Firestore data writes and reads obey owner-only security rules.
- Confirm responsive behavior and full-screen mobile experience remain stable.
- Confirm release readiness through successful build and deployment checks.

2. Test Strategy and Methodology

2.1 Test levels

- Component-level checks for key reusable widgets and auth UI interactions.
- Integration checks for AppContext actions and Firebase operations.
- End-to-end manual scenarios from registration to onboarding and dashboard.
- Deployment-level sanity checks on live production endpoint.

2.2 Test design techniques

Technique	Application in FitPulse	Expected Benefit
Equivalence partitioning	Email/password and profile input classes	Balanced coverage with reduced redundancy
Boundary value analysis	Numeric profile metrics and notifications	Early detection of threshold defects
Decision table testing	Verified vs unverified login outcomes	Deterministic validation of auth branches
State transition testing	Auth, onboarding, workout lifecycle transitions	Prevents illegal state progressions
Risk-based testing	Prioritization of auth and persistence blockers	Fast mitigation of high-impact defects

2.3 Risk-based prioritization

Risk Area	Likelihood	Impact	Priority
Auth flow failure or loop	Medium	High	Very High
Firestore access denial	Medium	High	Very High
Missing environment configuration	Medium	High	Very High
Google popup interruption	Medium	Medium	High
Responsive UI regression	Low	Medium	Medium

3. Test Environment and Data Setup

Item	Configuration
OS	Windows 11
Runtime	Node.js and npm scripts
Firebase Project	fitpulse-7173f
Environment Config	VITE_FIREBASE_* values via .env and Vercel variables
Build command	npm run build
Deployment endpoint	https://fitpulse-swe.vercel.app/
Baseline commit	8f7f44f

4. Test Case Specification

4.1 Authentication and account lifecycle

ID	Scenario	Steps	Expected Result
TC-AUTH-01	Register with valid email/password	Create account from auth page	Verification email sent and user signed out
TC-AUTH-02	Login with unverified email	Attempt sign-in before verification	Access blocked with verification message
TC-AUTH-03	Login with verified email	Sign in using valid verified credentials	Session restored and protected flow accessible

ID	Scenario	Steps	Expected Result
TC-AUTH-04	Forgot password flow	Submit reset request	Password reset email is dispatched
TC-AUTH-05	Google popup success	Continue with Google and complete consent	User session is created successfully
TC-AUTH-06	Google popup blocked	Block popup and retry Google sign-in	Redirect fallback is triggered
TC-AUTH-07	Logout behavior	Sign out from protected page	State resets and app returns to landing

4.2 Firestore and profile synchronization

ID	Scenario	Expected Result
TC-DATA-01	First login profile creation	User document is created with uid-scoped data
TC-DATA-02	Subsequent login merge	Existing profile is merged without field loss
TC-DATA-03	Onboarding persistence	Selections and stats are saved to Firestore
TC-DATA-04	Profile edit sync	Updated profile values are visible after refresh
TC-DATA-05	Cross-user data isolation	User cannot read or write another uid document

4.3 UI, state transitions, and responsiveness

ID	Scenario	Expected Result
TC-UI-01	Landing to auth navigation	Auth page renders without state corruption
TC-UI-02	Onboarding completion routing	Dashboard becomes default post-onboarding route
TC-UI-03	Workout start and end transitions	Timestamps and history are updated correctly
TC-UI-04	Mobile full-screen viewport	No shell gap or clipping on common mobile sizes
TC-UI-05	Notification state changes	Badge updates correctly for zero and non-zero values

4.4 Build and deployment validation

ID	Check	Expected Result
TC-REL-01	Run production build	Build succeeds with generated dist artifacts
TC-REL-02	Validate Firebase env keys in runtime	No missing-variable runtime error
TC-REL-03	Deploy Firestore rules	Rules compile and release successfully
TC-REL-04	Open production URL	Landing, auth, and protected pages load correctly

5. State-based Test Coverage

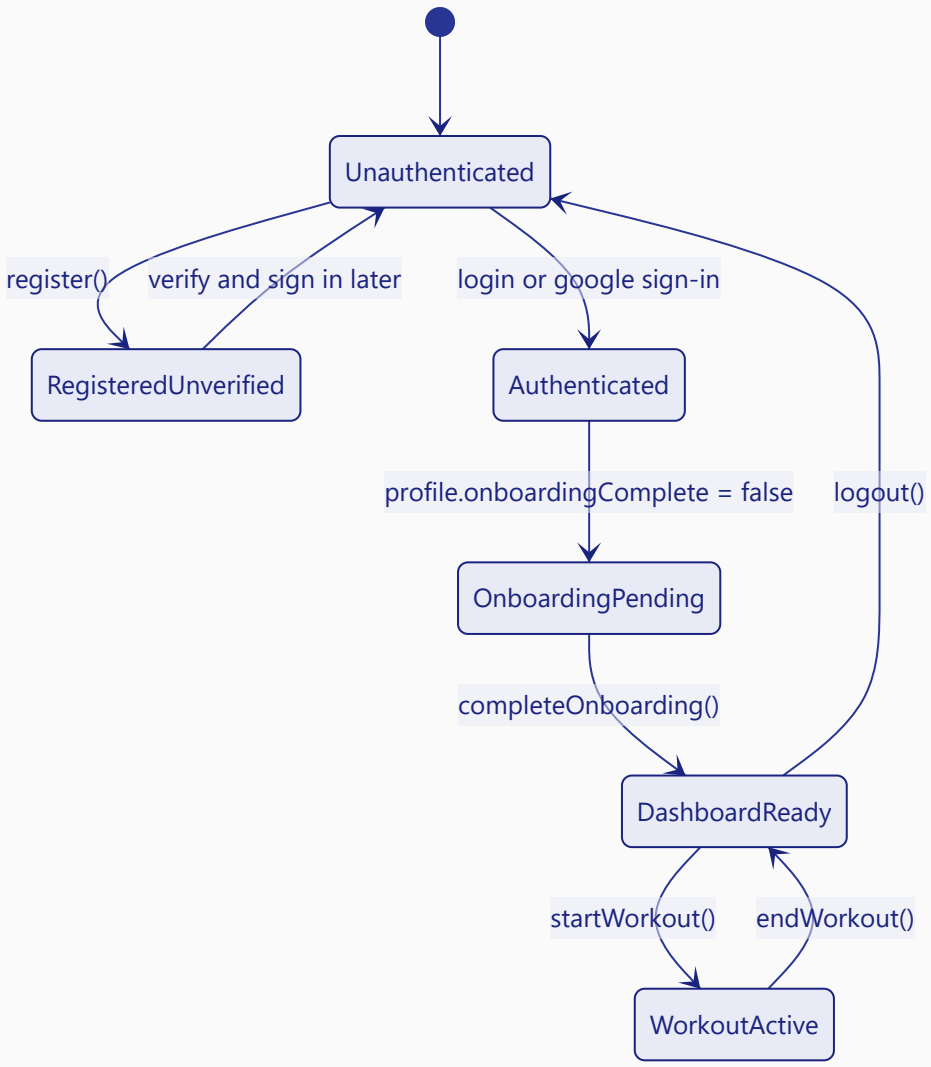

Figure 1: State model used for transition-oriented testing

6. Requirements Traceability Matrix

Requirement ID	Requirement	Mapped Cases	Status
FR-01	User authentication and routing	TC-AUTH-01 to TC-AUTH-07	Covered

Requirement ID	Requirement	Mapped Cases	Status
FR-02	Profile data persistence	TC-DATA-01 to TC-DATA-04	Covered
FR-03	Data isolation by user identity	TC-DATA-05	Covered
FR-04	State transition correctness	TC-UI-02, TC-UI-03	Covered
NFR-01	Responsive UI behavior	TC-UI-04	Covered
NFR-02	Release readiness	TC-REL-01 to TC-REL-04	Covered

7. Defect Classification and Reporting Format

Severity	Definition	Target Resolution
Critical	Complete flow blocker, auth crash, or data-access failure	Immediate patch
High	Major feature malfunction without reliable workaround	Within 24 hours
Medium	Partial functionality issue with workaround	Within sprint cycle
Low	Cosmetic inconsistency without functional impact	Next refinement cycle

> **Defect entry format:**
> Defect ID | Module | Build/Environment | Steps to reproduce | Actual result | Expected result | Severity | Status | Assignee

8. Test Metrics and Exit Criteria

Metric	Target	Observed Status
Critical flow pass rate	100%	Achieved for auth, routing, and state transitions
High-priority case pass rate	>= 95%	Achieved in current baseline
Build verification	Mandatory pass	Production build successful

Metric	Target	Observed Status
Production smoke checks	No blocking defects	No open blocker in final validation set

Exit criteria: all critical and high-priority scenarios pass, deployment checks pass, and no unresolved critical defects remain in the release candidate.

9. Deployment and Repository Links

Resource	Link	Usage
Live Production	https://fitpulse-swe.vercel.app/	Runtime and UX verification
GitHub Repository	https://github.com/abhinavsai2006/fitpulse	Traceability to implementation and revision history

10. Conclusion

The updated test design now reflects full production-aligned validation for FitPulse, including account verification behavior, cloud data security checks, and deployment confidence gates. This report provides a complete and professional test basis for academic submission.

End of Document

Designing Test Cases Report - FitPulse
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